

Translation

The following plan from 2018 is one of China's earliest national AI industrial policies. The plan identifies 16 specific application areas for AI technology, including vehicles, robots, UAVs, medical devices, IoT, AI chips, smart manufacturing, and cybersecurity, and sets goals for each of these to reach by 2020. The plan includes forms and instructions to guide Chinese companies in applying for government support to develop these technologies.

Title

Work Plan Unveiling the Key Tasks for Innovation in the New Generation Artificial Intelligence Industry

新一代人工智能产业创新重点任务揭榜工作方案

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Source

Website of the Central People's Government of the People's Republic of China (中国政府网). The Work Plan is dated November 8, 2018 and was uploaded to the website on December 31, 2018.

The Chinese source text is available online at:

<https://www.gov.cn/zhengce/zhengceku/2018-12/31/5443931/files/c6c6415dc9314119b488873f6e776ea1.doc>

An archived version of the Chinese source text is available online at: <https://perma.cc/78EK-QRNQ>

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Work Plan Unveiling the Key Tasks for Innovation in the New Generation Artificial Intelligence Industry

This Plan is formulated to implement the requirements of the *New Generation Artificial Intelligence Development Plan*¹ ([2017] State Council document no. 35) and the *Three-Year Action Plan for Promoting Development of a New Generation Artificial Intelligence Industry (2018–2020)*² ([2017] MIIT document No. 315), and to accelerate

¹ Translator's note: For an English translation of the New Generation Artificial Intelligence Development Plan, see: <https://www.newamerica.org/cybersecurity-initiative/digichina/blog/full-translation-chinas-new-generation-artificial-intelligence-development-plan-2017/>.

² Translator's note: For an English translation of the Three-Year Action Plan for Promoting Development of a New Generation Artificial Intelligence Industry (2018-2020), see:

innovation-driven development of China's new generation artificial intelligence (AI) industry.

I. Work Objectives

Focusing on the key directions of “cultivating intelligent products, making breakthroughs in core foundations, deepening the development of smart manufacturing, and building supporting systems,” this Plan will solicit and select a number of work units that have mastered key and core technologies (关键核心技术) and possess strong innovation capacity to carry out concentrated efforts on major challenges. The aim is to achieve breakthroughs in a batch of technologically advanced, high-performance iconic AI products, platforms, and services with fruitful applications; establish benchmarks and direction for innovation across the industry; and foster the main forces driving innovation in China's AI industry.

II. Challenge Tasks (揭榜任务) and Expected Targets

(i) Intelligent Products

1. Intelligent Connected Vehicles

Challenge task: Conduct R&D on key technologies and products including autonomous driving chips, intelligent algorithms for vehicles, autonomous driving systems, and onboard communications systems. Build a vehicle intelligentization (智能化) platform centered on a vehicle intelligent computing platform that integrates software, hardware, algorithms, connected communications, and information security.

Expected targets: By 2020, achieve breakthroughs in key technologies such as intelligent chips for autonomous driving, intelligent algorithms for vehicles, autonomous driving, and onboard communications. Enable intelligent connected vehicles to reach the level of autonomous driving with preconditions (有条件自动驾驶). Narrow the gap with the international advanced level in intelligent chips for autonomous driving in computing capabilities such as image processing, information fusion, and intelligent control. Bring the communication capacity and energy-efficiency ratio of onboard V2X systems up to the international advanced level. Meet the computing and communication technical requirements for intelligent perception, autonomous decision-making, cooperative control, and intelligent information exchange and sharing under autonomous driving with preconditions. Complete technical and functional verification and application demonstrations of a safe and

<https://www.newamerica.org/cybersecurity-initiative/digichina/blog/translation-chinese-government-outlines-ai-ambitions-through-2020/>.

reliable vehicle intelligentization platform. Form platform-related standards to support highly automated (HA-level) driving.

2. Intelligent Service Robots

Challenge task: Carry out R&D on key technologies including intelligent interaction, intelligent operation, multi-robot collaboration, 3D imaging and positioning, intelligent, precise, and safe control, and human-machine collaboration interfaces. Develop home service robots such as those for cleaning, elder companionship, rehabilitation, assistance for persons with disabilities, and children's education; public service robots such as inspection and tour-guide robots; special service robots such as fire-fighting and rescue robots; and surgical robots and their operating systems.

Expected targets: By 2020, achieve breakthroughs in key technologies such as environmental perception, natural interaction, autonomous learning, and human-machine collaboration for intelligent service robots. Realize mass production and application of smart home service robots and intelligent public service robots. Complete prototype development of robots for medical rehabilitation, elder and disability assistance, and fire and disaster rescue. Complete technical and functional verification and application demonstrations.

3. Smart UAVs

Challenge task: Conduct R&D and application of key technologies including intelligent obstacle avoidance, automatic cruising, autonomous flight in complex environments, and swarm operations. Apply next-generation communication, positioning, and navigation technologies in UAV data transmission, link control (链路控制), and monitoring and management. Develop key components such as intelligent flight control systems and highly integrated dedicated chips.

Expected targets: By 2020, achieve a precision of 0.005 degrees for the three-axis mechanical gimbal stabilizers of intelligent consumer-grade UAVs. Realize 360-degree omnidirectional perception and obstacle avoidance, achieve automatic and intelligent forced avoidance of air traffic control zones, and develop products with strong global competitiveness.

4. Medical Imaging-Assisted Diagnosis Systems

Challenge task: Standardize and regulate medical imaging data collection; conduct R&D on medical imaging-assisted diagnostic technologies for typical disease areas such as the brain, lung, eye, bone, cardiovascular and cerebrovascular system, and breast; and develop products and clinical assistance applications of medical

imaging–assisted diagnosis systems.

Expected targets: By 2020, have multimodal medical imaging–assisted diagnosis systems achieve detection rates exceeding 95% for typical diseases of the brain, lung, eye, bones, cardiovascular and cerebrovascular systems, and breast. False-negative rates should be below 1%, false-positive rates below 5%, and clinical applications should gradually expand.

5. Video and Image Identity Recognition Systems

Challenge task: Innovate technologies including biometric feature recognition, video understanding, and cross-media integration. Develop typical applications such as identity verification, video surveillance, image search, and video summarization.

Expected targets: By 2020, in complex and dynamic scenarios, achieve facial recognition effective detection rates (有效检出率) exceeding 97% and correct recognition rates (正确识别率) exceeding 90%. Support recognition of facial features across different regions, and develop products that reach the international advanced level.

6. Intelligent Speech Interaction Systems

Challenge task: Innovate technologies including new-generation speech recognition frameworks, colloquial speech recognition, personalized speech recognition, intelligent dialogue, audio-video integration, and speech synthesis, and promote their application in key fields such as smart manufacturing and smart homes.

Expected targets: By 2020, achieve an average accuracy rate of 96% for Chinese speech recognition across multiple scenarios, exceed 92% recognition accuracy for far-field speech at 5 meters, and surpass 90% accuracy in recognizing user dialogue intentions. Develop products that reach the international advanced level.

7. Intelligent Translation Systems

Challenge task: Innovate high-precision intelligent translation systems and apply them in typical scenarios such as multilingual translation and simultaneous interpretation.

Expected targets: By 2020, achieve significant breakthroughs in multilingual intelligent translation, with translation accuracy exceeding 85% for Chinese-to-English and English-to-Chinese scenarios. Substantially improve the accuracy of intelligent translation between ethnic minority (少数民族) languages and Chinese, and develop products that reach the international advanced level.

8. Smart Home Products

Challenge task: Integrate and apply technologies such as intelligent sensing, the internet of things (IoT), and machine learning in products such as smart security, smart furniture, smart lighting, and smart bathroom fittings. Develop new smart home products.

Expected targets: By 2020, significantly enhance the level of intelligentization of smart home products such as smart security and smart appliances, and achieve strong demonstrative applications of such products.

(ii) Core Foundations

9. Smart sensors

Challenge task: Conduct R&D on key technologies including miniaturization and reliability design, precision manufacturing, integrated development tools, and embedded algorithms. Develop and apply smart sensors designed around new requirements, new materials, new processes, and new principles. Carry out R&D and application of new types of smart sensors for biological, gas, pressure, flow, inertial, distance, imaging, and acoustic sensing. Innovate in materials technologies including piezoelectric materials, magnetic materials, infrared radiation materials, and metal oxides. Develop new smart sensors based on processes such as micro-electromechanical systems (MEMS) and complementary metal-oxide-semiconductor (CMOS) integration. Develop new smart sensors for emerging application scenarios based on new principles such as magnetic sensing, ultrasound, non-visible light, and biochemistry.

Expected targets: By 2020, significantly improve the performance of piezoelectric sensors, magnetic sensors, infrared sensors, and gas sensors. Achieve a signal-to-noise ratio (SNR) of 70 dB for acoustic sensors and an acoustic overload point (AOP) of 135 dB, with mass production capability. Achieve an absolute precision (绝对精度) for pressure sensors within 100 Pa and a noise level within 0.6 Pa, with commercialization capability. Achieve a weak magnetic field resolution of 1 pT for magnetic sensors, with mass production capability. Reach the international advanced level in simulation, design, MEMS processes, packaging, and customized testing technologies.

10. Neural Network Chips

Challenge task: Develop cloud-based neural network chips for machine learning training applications and terminal device (终端) neural network chips suitable for machine learning computation in end-use applications. Develop commercialization support tools to accompany neural network chips, including compilers, driver software, and development environments.

Expected targets: By 2020, have cloud-based neural network chips achieve performance of 128 TFLOPS (16-bit floating point) with an energy efficiency ratio exceeding 1 TFLOPS/W, and have terminal device neural network chips achieve an energy efficiency ratio exceeding 1 TOPS/W (based on 16-bit floating point). Support one or more mainstream neural network algorithms such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and long short-term memory networks (LSTMs). Realize large-scale commercialization of neural network chips in key areas such as intelligent terminal devices, autonomous driving, intelligent security, and smart homes.

11. Open-Source (开源) and Open (开放) Platforms

Challenge task: Develop development frameworks, algorithm libraries, and toolkits for cloud-based training and terminal device execution. Build open-source development platforms, open technology networks, and open-source communities, as well as open computing service platforms capable of meeting complex training needs.

Expected targets: By 2020, open-source development platforms for cloud-based training will support large-scale distributed clusters (分布式集群), multiple hardware platforms, and multiple algorithms. Open-source development platforms for execution on terminal devices will be lightweight, modular, and reliable.

(iii) Smart Manufacturing Key Technologies and Equipment

12. Smart Manufacturing Key Technologies and Equipment

Challenge task: Carry out R&D and application of industrial robots with self-detection, self-calibration, self-adaptation, and self-organization capabilities. Innovate and apply control equipment such as smart sensors and distributed control systems (DCS), programmable logic controllers (PLC), supervisory control and data acquisition systems (SCADA), and high-performance, highly reliable embedded control systems. Develop and apply intelligent inspection equipment based on AI technologies such as image recognition and deep learning. Innovate and apply logistics equipment such as high-speed sorting machines, multi-level pallet shuttles (多层穿梭车), and high-density storage shuttles (高密度存储穿梭板) based on AI technologies.

Expected targets: By 2020, new-generation industrial robots will achieve functions such as human-machine coordination, natural interaction, and autonomous learning, and realize mass production and application. Smart sensing and control equipment will achieve integrated application in fields such as machine tools, robotics, petrochemicals, and rail transit. Intelligent inspection and assembly equipment will achieve 90% accuracy in visual recognition in industrial settings, with measurement precision and speed meeting actual production requirements. Intelligent logistics and

warehousing equipment will achieve a significant improvement in intelligentization, meeting the needs of precision, flexible, and efficient material distribution and dark warehousing.

(iv) Supporting Systems

13. Industry Training Resource Libraries

Challenge task: Build high-quality AI training resource libraries, standard test datasets, and cloud service platforms to support R&D on key products such as speech recognition, visual recognition, and natural language processing, as well as applications in industries such as manufacturing, healthcare, finance, and transportation.

Expected targets: By 2020, substantially increase the volume of public training data for basic speech, video and images, and text dialogue. Aggregate a sizable amount of industry application data in manufacturing, healthcare, finance, and transportation to support entrepreneurship and innovation.

14. Standards Testing and Intellectual Property Service Platforms

Challenge task: Establish a standards and regulatory system for the AI industry covering technologies such as basic and general purpose technologies, interoperability, security and privacy, and industry applications. Build AI product evaluation and testing systems, collaborative patent operation platforms, and intellectual property service platforms.

Expected targets: By 2020, preliminarily establish an AI industry standards system. Build third-party pilot testing platforms and carry out evaluation and testing services. In fields such as pattern recognition, semantic understanding, autonomous driving, and intelligent robots, establish intellectual property service platforms with foundational support capabilities.

15. Intelligentized Network Infrastructure

Challenge task: Develop highly intelligentized next-generation internet, high-speed, large-capacity, low-latency 5G networks, navigation networks with fast and high-precision positioning, and ubiquitous, integrated, and efficient space-ground information networks. Build intelligentized network infrastructure such as the industrial internet and the internet of vehicles.

Expected targets: By 2020, more than 90% of regions nationwide will achieve broadband access speeds and latency levels sufficient to meet the needs of AI industry applications. Industrial internet demonstration projects covering the entire production

process will have been implemented in key enterprises. Initial internet of vehicles network infrastructure will be established in key regions.

16. Cybersecurity Assurance Systems

Challenge task: Meet the security needs of key AI products and industry applications such as intelligent connected vehicles and smart homes, and drive innovation in cybersecurity products based on advanced AI technologies.

Expected targets: By 2020, form a framework for an AI security [risk] prevention and control system. Build an initial security assurance platform with basic capabilities for AI security situational awareness, testing and evaluation, threat information sharing, and emergency response.

17. Other

Other important technologies, products, services, and platforms involved in the development of the AI industry.

III. Implementation Steps

(i) Application for challenge tasks.

Enterprises, universities, research institutes, and other legal entities engaged in AI technology innovation, product R&D, integrated applications, or supporting services, as well as consortia formed by multiple entities, may apply to become challenge task units. Applicants should possess strong innovation capabilities, hold intellectual property rights to the products or services submitted for the challenge, and demonstrate technological advancement and good application prospects. Applicants must commit to completing the challenge task within the specified time frame after acceptance.

(ii) Recommendation of units.

The industry and information technology departments (hereafter collectively referred to as the “main oversight departments”) of provinces, autonomous regions, province-level municipalities, cities with independent planning status under the national economic and social development plan,³ and the Xinjiang Production and Construction Corps; central enterprise⁴ groups; and AI-related industry organizations

³ Translator's note: The five Chinese "cities with independent planning status under the national economic and social development plan" (国家社会与经济发展计划单列市; 计划单列市) are Dalian, Qingdao, Ningbo, Xiamen, and Shenzhen.

⁴ Translator's note: The term "central enterprises" (中央企业; 央企) refers to state-owned enterprises (SOEs) directly under the Central People's Government of the People's Republic of China (aka the State

serve as recommending units. They will organize relevant enterprises, universities, and research institutes to complete application materials, review them, and then submit recommendation forms and hardcopy application materials in a consolidated manner.

(iii) Selection of challenge task units.

The Ministry of Industry and Information Technology (MIIT) will organize industry experts and evaluation institutions to conduct centralized reviews and on-site assessments. Taking into account each applicant's baseline capabilities, innovation capacity, development potential, product indicators, and regional industrial layout, MIIT will select the best candidates and publish the list of challenge task units (in principle, no more than five per sub-field).

(iv) Implementation of challenge tasks.

Challenge task units shall organize and implement their assigned tasks as required and carry out concentrated efforts on major challenges. During this period, MIIT will continuously track the progress of product innovation and application by the challenge task units. MIIT will, when appropriate, organize industry experts to conduct phased (阶段性) evaluations of the challenge tasks and will continuously optimize the challenge task implementation pathways.

(v) Publication of challenge results.

Upon completing their research tasks, challenge task units may apply for evaluation. Based on progress, MIIT will organize industry experts and entrust qualified third-party professional institutions with the necessary testing and evaluation work. Evaluation will be conducted in accordance with the challenge tasks and expected targets, combined with the actual circumstances of industrial development. Results will be announced in due course, and units that have successfully completed the tasks will be selected (in principle, no more than three per sub-field) for priority promotion and support.

IV. Work Requirements

The local main oversight departments and central enterprise groups shall strengthen organizational leadership and fully mobilize the initiative of enterprises, research institutes, and relevant industry alliances and industry associations. They

Council) whose leaders are appointed by the State Council or the Chinese Communist Party (CCP) Central Committee, either directly or by the CCP Central Organization Department, the State-owned Assets Supervision and Administration Commission (SASAC), or another central Party or government department.

shall closely track the progress of product innovation and application by challenge task units, conduct phased evaluations of challenge tasks when appropriate, and effectively coordinate and advance the organization and implementation of concentrated research efforts. Localities are encouraged to align with the development of their regional AI industries and give priority support in terms of related matching funding, projects, and other measures, thereby creating a favorable environment for challenge task units to accomplish their research tasks.

Appendixes:

1. Challenge Task Unit Recommendation Form for Key Innovation Tasks in the New Generation AI Industry
2. Challenge Task Unit Application Materials for Key Innovation Tasks in the New Generation AI Industry (Template)

Appendix 1

Challenge Task Unit Recommendation Form for Key Innovation Tasks in the New Generation AI Industry

Recommending Unit (official seal):

No.	Organization Name	Direction	Challenge Products and Services	Reason for Recommendation	Contact Person	Mobile No.
1						
2						
3						

Notes: 1. This form shall be completed by recommending units such as local authorities, central enterprises, alliances, or associations

2. Recommending units shall rank the recommendations in order of priority

3. "Direction" (所属方向) refers to the 17 key task directions listed for the challenge tasks and expected targets

Appendix 2

Challenge Task Unit Application Materials for Key Innovation Tasks in the New Generation AI Industry

Challenge Direction:_____

Sub-field:_____

Applicant Unit:_____(official seal)_____

Recommending Unit:_____(official seal)_____

Date of Application:_____(MM/DD/YYYY)

Instructions

1. Challenge task applicant units should carefully read the relevant instructions in the *Work Plan Unveiling the Key Tasks for Innovation in the New Generation AI Industry* and fill in each part truthfully and in detail.

2. Unless otherwise specified, no fields in the application form may be left blank. Where supporting materials are required, please attach the corresponding documents.

3. For hardcopy application materials, all required seals must be affixed; photocopies are not valid. The application materials must carry a cross-page seal (骑缝章) and be submitted together with the corresponding supporting materials to the recommending unit for mailing.

4. The content and format of the electronic materials must be consistent with the hardcopy materials. In case of discrepancies, the hardcopy materials shall prevail.

5. The applicant must possess intellectual property rights to the submitted product. The applicant is responsible for the authenticity of all submitted materials and must make a valid commitment to completing the key challenge task as planned by signing an Enterprise Commitment Statement (see the Challenge Task Commitment Letter template).

Challenge Task Unit Application Form for Key Innovation Tasks in the New Generation AI Industry

I. Organization Information (provide supporting documents)				
Organization Name	Full Name (fill in truthfully)			
Challenge Lead	Name		Title/Position	
	Email		Mobile No.	
Application Contact Person	Name		Mobile No.	
	Email		Fax	
Legal Representative			Registered Capital	
Organization Address				
Organization Code / Three-in-One License Number ⁵				
Unit Type	☐ State-owned enterprise ☐ Private enterprise ☐ Foreign-funded enterprise ☐ Public institution ⁶ Other (please specify): _____			
Publicly Listed?	☐ No ☐ Yes (Stock Code: _____ Listing Location: _____ Listing Date: _____)			
Total Business Revenue	Refers to the previous fiscal year (provide proof)		R&D Investment	Refers to the previous fiscal year (provide proof)
Revenue from AI-related Business				
Total Employees			Number of R&D Personnel	

⁵ Translator's note: In China, a business' "three-in-one license" (三证合一) is a single license with a single license number that serves simultaneously as a commercial operating license (营业执照), a certificate affirming the company's organization code number (组织机构代码证), and tax registration certificate (税务登记证). These were once three separate documents that required three separate licensing processes, until a mid-2010s reform allowed for them to be combined into one.

⁶ Translator's note: "Public institutions" (事业单位) are organizations created and led by Chinese government departments that perform a variety of functions. Notably, universities, research institutes, and hospitals in China tend to be public institutions. Unlike state-owned enterprises (SOEs), public institutions do not create material products and are non-profit. Public institutions are not considered government agencies, and their employees are not civil servants. Most public institutions are fully or partially government-funded, but some fully privately funded (but still government-led) public institutions exist.

Brief Introduction to the Applicant for the Challenge Task	Include basic information such as founding date, main business areas, main products, technological capabilities, and development history, as well as relevant patents, standards, intellectual property rights, and awards from competitions (Attach relevant supporting documentation below) (This section shall not exceed 500 words)
Participating Units	
II. Basic Information on Key Products	
Name of Challenge Product	
Key Product Direction	Intelligent Products ☐ Direction 1: Intelligent Connected Vehicles ☐ Direction 2: Intelligent Service Robots ☐ Direction 3: Smart UAVs ☐ Direction 4: Medical Imaging-Assisted Diagnosis Systems ☐ Direction 5: Video and Image Identity Recognition Systems ☐ Direction 6: Intelligent Speech Interaction Systems ☐ Direction 7: Intelligent Translation Systems ☐ Direction 8: Smart Home Products Core Foundations ☐ Direction 9: Smart Sensors ☐ Direction 10: Neural Network Chips ☐ Direction 11: Open-Source and Open Platforms Smart Manufacturing Key Technologies and Equipment ☐ Direction 12: Smart Manufacturing Key Technologies and Equipment Supporting Systems ☐ Direction 13: Industry Training Resource Libraries ☐ Direction 14: Standards Testing and Intellectual Property Service Platforms ☐ Direction 15: Intelligentized Network Infrastructure ☐ Direction 16: Cybersecurity Assurance Systems Other ☐ Direction 17: Other
Overview of the Challenge Product	Include a description of the product/service submitted for the challenge, overview of investment and financing, relevant R&D and application maturity, and the expected technical and commercial application level to be achieved by 2020 (If multiple products are involved across different domains, please describe them separately) (Limit: 1,000 words)

Key Indicators Declaration Form for Products or Services Submitted by Challenge Task Applicant Units

Challenge Direction	Segments	Challenge Product	Reference Indicators	Current Level of the Unit	Unit's 2020 Target	Supplementary Explanation of Benchmark Scenarios or Specific Meanings of Indicator Levels
Example: Intelligent Connected Vehicles	Example: Autonomous Driving Operating System		Autonomous driving level			
			Interventions per kilometer			
			Distance already road tested			
			Intelligent environmental perception field of view, detection range, resolution, etc.			
			Precision planning and control capability for autonomous decision-making			
			High-precision map data collection and service capability			
			Number of vehicle models supported			
			Categories and number of V2X applications supported			
			Number of autonomous driving application scenarios supported			
			Commercialization and application status			
			Other indicators...			

Notes: 1. The indicators in this table mainly include technical performance indicators and commercialization indicators. They are not disclosed publicly and are used only as reference for review by experts and evaluation institutions.

2. Indicators submitted by challenge task units must include those mentioned in the "Challenge Tasks and Expected Targets" section. Additional reasonable indicators may also be added. The "Unit's 2020 Target" column should at minimum reflect the lower bound of the expected achievable indicator level. Submitting targets that exceed the 2020 targets set in the *Three-Year Action Plan for Promoting Development of a New Generation Artificial Intelligence Industry (2018-2020)* is encouraged.

Directory of Key Tasks and Reference Indicators for Products or Services Submitted by Challenge Task Applicant Units

Challenge Direction	Segments	Reference Indicators
1.Intelligent connected vehicles	Autonomous driving operating system	Autonomous driving level
		Interventions per kilometer
		Distance already road tested
		Intelligent environmental perception field of view, detection range, resolution, etc.
		Precision planning and control capability for autonomous decision-making
		High-precision map data collection and service capability
		Number of vehicle models supported
		Categories and number of V2X applications supported
		Number of autonomous driving application scenarios supported
		Commercialization and application status
		Other indicators...
	Smart chips for autonomous driving	Support for autonomous driving functions and tasks
		Visual information processing capability
		Performance/power consumption
		Does it have an independent instruction set/independent IP?
		Per-frame latency
		Safety and security
		Commercialization and application status
		Other indicators...
	On-board communication systems	Communication capability of on-board V2X information networking systems
		Performance of on-board communication chips
		Categories and number of V2X applications supported
		Level of compliance with standards
		Communication security assurance capability
		Commercialization and application status
		Other indicators...
2.Intelligent service robots	Smart home service robots	Intelligent interaction capability
		Autonomous action capability
		Scale of intelligent service knowledge base and quality of knowledge representation
		Safety and security
		Commercialization and application status
		Other indicators...

Challenge Direction	Segments	Reference Indicators
	Intelligent education robots	Intelligent interaction capability
		Bionic action capability
		Scale of intelligent service knowledge base and quality of knowledge representation
		Commercialization and application status
		Other indicators...
	Intelligent public service robots	Environmental perception capability
		Autonomous obstacle-avoidance capability
		Intelligent interaction capability
		Scale of intelligent service knowledge base and quality of knowledge representation
		Safety and security
		Commercialization and application status
		Other indicators...
	Intelligent specialized robots	Intelligent perception capability
		Autonomous decision-making capability
		Human-machine collaboration capability
		Autonomous obstacle-avoidance capability
		Autonomous navigation capability
		Reliability
		Commercialization and application status
		Other indicators...
	Intelligent surgical robots	3D imaging and positioning capability
		Intelligent, precise, and safe control capability
		Human-machine collaboration capability
		Reliability
		Commercialization and application status
		Other indicators...
	Intelligent customer service robots	Coverage of business scenarios
		Accuracy of intent recognition
		User problem resolution rate
		Scale of knowledge base and quality of knowledge representation
		Number of large-scale deployment cases/number of dialogues handled
		Other commercialization and application status
		Other indicators...
3.Smart UAVs	Smart UAVs	Perception and obstacle-avoidance capability
		Gimbal stabilizer precision
		Automatic forced avoidance capability for air traffic control zones
		Intelligent swarm operation capability
		Data transmission communication rate
		Commercialization and application status

Challenge Direction	Segments	Reference Indicators
		Other indicators...
4. Medical imaging-assisted diagnosis systems	Medical imaging-assisted diagnosis systems	Detection rate for typical diseases
		False-negative rate for typical diseases
		False-positive rate for typical diseases
		Amount of existing clinical imaging data in the system
		Commercialization and application status
		Other indicators...
5. Video and image identity recognition systems	Video surveillance and security identity recognition systems	Effective detection rate for facial recognition in complex and dynamic scenarios
		Size of registration set supported for 1:N facial recognition in complex and dynamic scenarios
		Correct recognition rate (1:N) in complex and dynamic scenarios
		Support for recognition of facial features across different regions
		Support for gait recognition and other recognition methods
		System response time
		Commercialization and application status
		Other conditions
	Smart terminal device facial recognition systems	Effective detection rate for facial recognition in complex and dynamic scenarios
		Correct recognition rate in complex and dynamic scenarios
		Support for recognition of facial features across different regions
		System resource usage
		System response time
		Liveness detection (活体检测) / anti-spoofing (防攻击) capability
		Commercialization and application status
		Other conditions
6. Intelligent speech interaction systems	Intelligent speech interaction systems	Average accuracy/recall rate of Chinese speech recognition across multiple scenarios
		Far-field recognition accuracy at 5 meters
		Accuracy of user intent recognition in multi-round conversation
		Types of international languages, ethnic minority languages, and dialects supported
		Wake-up rate/false wake-up rate
		Average response time
		Service satisfaction rate and accurate recommendation capability
		Cross-device user experience for members of the same household
		Commercialization and application status

Challenge Direction	Segments	Reference Indicators
		Other indicators...
7.Intelligent translation systems	Intelligent translation systems	Language pairs supported for two-way intelligent translation
		Translation accuracy for Chinese-to-English and English-to-Chinese scenarios
		Accuracy of intelligent translation between ethnic minority languages and Chinese
		Translation response time
		Maximum simultaneous translation (并发翻译) capacity
		Support for and accuracy of offline translation
		Translation accuracy for speech/image input
		Readability (fluency)
		Daily request volume
		Other commercialization and application status
		Other indicators...
8.Smart home products	Smart security	Comprehensive perception and recognition capability for multiple environmental factors
		Intelligent risk warning capability
		Terminal device data computation capability
		Cloud-based massive data processing capability
		Safety and security
		Commercialization and application status
		Other indicators...
	Smart appliances	Whether intelligent sensing components are embedded, and their number and type (e.g., motion sensors, microphones)
		Whether intelligent chips are embedded, and their number and type
		Intelligent interaction capability
		Capability to provide personalized intelligent services through learning
		Intelligent management capability
		Intelligent malfunction prediction capability
		Safety and security
		Commercialization and application status
		Other indicators...
9.Smart sensors	Smart sensors	Sensitivity
		Precision
		Resolution
		Degree of intelligent processing of data on the sensor side
		Power consumption
		Commercialization and application status
		Other indicators...
10.Neural network chips	Cloud-based neural network chips	Number and types of mainstream neural network algorithms supported
		Performance (16-bit floating point)

Challenge Direction	Segments	Reference Indicators
		Performance/power consumption
		Independent instruction set/independent IP
		High-performance interconnection between chips
		Commercialization and application status
		Other indicators...
	Terminal device neural network chips	Number and types of mainstream neural network algorithms supported
		Performance/power consumption (on 16-bit floating point basis)
		Independent instruction set/independent IP
		Commercialization and application status
		Other indicators...
11.Open-source and open platforms	Open-source and open platforms	Large-scale distributed cluster parallel processing capability
		Support capability for different hardware platforms
		Support capability for multiple algorithm models, frameworks, and complex training tasks
		Number of active developers in the open-source community
		Types/number of applications developed and provided as services
		Number of datasets/models/users
		Application isolation methods and developer user experience
12.Key technologies and equipment for smart manufacturing	Intelligent industrial robots	Other
		Intelligent perception capability
		Human-machine collaboration capability
		Intelligent decision-making capability
		Flexible and precise control capability
		Abnormal situation handling capability
		Remote operation and maintenance capability
		Openness and capability for secondary development
		Commercialization and application status
		Other indicators...
	Intelligent control equipment	Intelligent perception capability
		Intelligent decision-making capability
		Intelligent precise control capability
		Commercialization and application status
		Other indicators...
	Intelligent inspection equipment	Visual recognition accuracy in industrial venues
		Measurement precision
		Processing speed
		Commercialization and application status
		Other indicators...
	Intelligent logistics	Positioning methods and precision
		Cargo classification and processing capability

Challenge Direction	Segments	Reference Indicators
	equipment	Level of human-machine collaboration
		Motion and route planning (运动路线规划) capability
		Agile obstacle-avoidance capability
		Operating cost reduction rate after application at scale
		Labor cost reduction rate after application at scale
		Commercialization and application status
		Other indicators...
13.Industry training resource libraries	Industry training resource libraries	Types/volume of data
		Industry sectors served
		Proportion and accuracy of labeled data
		Diversity of data distribution
		Openness and application status
		Other indicators...
14.Standards testing and intellectual property service platforms	Standards testing and intellectual property service platforms	AI standards service capability
		AI product evaluation capability
		Public service capability for intellectual property in the AI industry
		Standards testing and intellectual property service status
		Other indicators...
15.Intelligentized network infrastructure	Improvement in 4G/5G network infrastructure intelligentization	Scale of deployed network and number of nodes
		Average 4G/5G broadband access speed and latency
		Level of intelligentization in network performance fine-tuning and parameter optimization
		Capability for intelligent perception and monitoring of global network resources
		Intelligent error warnings, root-cause locating, and self-repair capability
		Security assurance capability for equipment, networks, and emergencies
		Platform service capability (e.g., APIs) to support typical AI applications such as autonomous driving, UAVs, robots, and large-scale video
		Number of typical AI applications already supported
		Other demonstration applications
		Other indicators...
	Industrial internet network infrastructure supporting industrial intelligentization applications	Scale of deployed network and number of nodes
		Identity resolution service capability
		Service provision capability for new technology applications such as NB-IoT, IPv6, SDN/NFV, time-sensitive networking (TSN), and edge computing
		Security assurance capability for equipment, networks, and emergencies
		Demonstration applications

Challenge Direction	Segments	Reference Indicators
	Internet of vehicles network infrastructure	Other indicators...
		Scale/coverage of deployed networks
		System capacity/connection density
		Bidirectional transmission speed and latency in typical application scenarios
		Support for multiple communication formats
		Security assurance capability for equipment, networks, and emergencies
		Demonstration applications
		Other indicators...
16.Cybersecurity assurance systems	AI security technology service platforms	Sample scale of vulnerability libraries, virus databases, etc.
		Types of intelligent products covered
		Capabilities for security testing, risk warning, and emergency response
		External service provision status
		Other indicators...
	Applications of AI in network, information, and data security fields	Improvement in security testing capability through application of AI technology
		Improvement in security protection capability through application of AI technology
		Improvement in the identification and handling of vulgar or harmful information through application of AI technology
		Improvement in data security storage, transmission, and processing capability through application of AI technology
		Commercialization and application status
		Other indicators...
	AI terminal device security protection systems	AI terminal device security protection capability
		Main control app security protection capability
		Capability to protect against leakage of private information
		Other indicators...
17.Other directions	Other sub-fields	Level of intelligentization
		Safety and security
		Commercialization and application status
		Other indicators

Notes: 1. If the above challenge task directions do not cover a given product, it may be entered under “17. Other directions, Other sub-fields.”

2. Commercialization and application status indicators include: economic and social benefits, market scale/market share, and demonstration applications.

Challenge Task Letter

I. Brief Introduction of the Challenge Task

Name of the product or service researched; main technologies involved; directions of innovation; development trends and prospects, etc.

II. Existing Capabilities and Relevant Progress of the Challenge Task

Applicant Unit

(1) Existing Capabilities

Industry position, scientific research qualifications, technological foundation, intellectual property rights, innovation capacity, strength of talent and teams, main advantages, and major events hosted/co-hosted/participated in.

(2) Relevant Progress

Current technical level of the key product or service researched (compared with the international advanced level), innovation and application status, related R&D personnel, investment of funds, etc.

III. Key R&D Objectives and Implementation Plan

(1) Expected Outcomes by 2020

Indicator values, their meaning, test scenarios, evaluation methods, etc.

(2) R&D Plan for Key Tasks

Include timelines, phased tasks, detailed objectives, etc.

(3) Organizational Support Mechanisms

R&D teams, organizational structure, coordination mechanisms, etc.

(4) Potential Challenges and Mitigation Measures

IV. Other Relevant Information

Note: In general, this application form should not exceed 6,000 words, with a focus on the objectives and implementation plan. If applying for projects in multiple fields, please submit separate applications based on this template for each field.

Supporting Documents of the Challenge Task Applicant Unit

1. Proof of main business revenue of the applicant unit in the previous fiscal year. (Financial accounting statements, tax certificates, etc.)

2. Proof of R&D investment of the applicant unit in the previous fiscal year.
(Financial accounting statements, etc.)
3. Proof of R&D capabilities of the applicant unit. (Patents obtained, standards, intellectual property, etc.)
4. Proof of honors and recognitions of the applicant unit. (Supporting documents proving status as a high-tech enterprise, enterprise technology center, or key laboratory, and awards received in competitions)
5. Proof of current performance indicators and application outcomes for the proposed product/service (Such as third-party testing reports)

Challenge Task Commitment Letter

According to the requirements of the *Notice on Issuing the Work Plan Unveiling the Key Tasks for Innovation in the New Generation AI Industry*, our unit hereby submits the

_____product for evaluation.

We hereby make the following commitments:

1. Our unit is responsible for the authenticity of all submitted materials. We guarantee that we hold intellectual property rights to the submitted products and application solutions and that the submitted products and services comply with the relevant national laws and regulations and with relevant policy requirements.
2. The products and services submitted by our unit comply with national secrecy protection regulations and do not involve state secrets, personal private data, or other sensitive information.
3. The text and images in the relevant materials have been reviewed by our unit and have been confirmed to be accurate.

Our entity shall bear full legal responsibility for any consequences arising from violations of the above commitments.

Our unit will, in accordance with the requirements of the Work Plan, strengthen awareness of the overall situation, and conscientiously assume primary responsibility. During the execution of the challenge tasks, we will carefully organize, focus on implementation, and enhance safeguards. We will make every effort to accomplish the key R&D objectives and strive to achieve substantive progress by 2020, reaching or exceeding the expected targets.

Contact person:

Telephone number:

Legal representative: (Signature)

Company (Official seal)

_____(month) __ (day), 2018